

Session:12

1. GST Lambda Function (18%)

```
gst = lambda price: price * 1.18

print(gst(100))
print(gst(250))
print(gst(500))
```

Output:

```
118.0
295.0
590.0
```

2. Clean Spotify Song Titles using map() + lambda

```
songs = [' shape OF you ', ' blinding LIGHTS ', '  senorita ', 'stay ']

cleaned = list(map(lambda s: s.strip().title(), songs))

print(cleaned)
```

Output:

```
['Shape Of You', 'Blinding Lights', 'Senorita', 'Stay']
```

3. Filter Flipkart Products Starting with 'S'

```
products = ['Shoes', 'bag', 'Smartphone', 'Sofa', 'table', 'Speaker']

filtered = list(filter(lambda x: x.lower().startswith('s'), products))

print(filtered)
```

Output:

```
['Shoes', 'Smartphone', 'Sofa', 'Speaker']
```

4. Total Bill using reduce()

```
from functools import reduce

orders = [120, 340, 560, 80]
```

```
total = reduce(lambda x, y: x + y, orders)
print(total)
```

Output:

```
1100
```

5. map() + filter() + reduce() together

Process: double → filter > 100 → sum

```
from functools import reduce

nums = [40, 60, 80, 120]

result = reduce(
    lambda x, y: x + y,
    filter(
        lambda x: x > 100,
        map(lambda x: x * 2, nums)
    )
)

print(result)
```

Step breakdown:

- Original: [40, 60, 80, 120]
- After map (×2): [80, 120, 160, 240]
- After filter (>100): [120, 160, 240]
- Sum: 520

Output:

```
520
```